Q1. JEE Main 2026 (21 January Shift 2)

If the area of the region $\{(x, y): 1-2x \leqslant y \leqslant 4-x^2, x \geqslant 0, y \geqslant 0\}$ is $\frac{\alpha}{\beta}, \alpha, \beta \in \mathbf{N}, \gcd(\alpha, \beta)=1$, then the value of $(\alpha+\beta)$ is :

(1) 67 (2) 91 (3) 73 (4) 85

Q2. JEE Main 2026 (23 January Shift 1)

Let the area of the region bounded by the curve $y=\max\{\sin x, \cos x\}$, lines $x=0, x=\frac{3\pi}{2}$, and the x-axis be A. Then, $\mathrm{A}+\mathrm{A}^2$ is equal to _____.

Q3. JEE Main 2026 (28 January Shift 2)

Let $P_1: y=4x^2$ and $P_2: y=x^2+27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y=\alpha x, \alpha>0$ and P_1, then α is equal to :

(1) 12 (2) 6 (3) 15 (4) 8

Q4. JEE Main 2026 (24 January Shift 2)

Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x=0, x=1, y^2=x$ and $y=|\alpha x-5|-|1-\alpha x|+\alpha x^2$. Then $(f(0)+f(1))$ is equal to

(1) 12 (2) 14 (3) 7 (4) 9

Q5. JEE Main 2026 (22 January Shift 2)

The area of the region $\mathrm{A}=\{(x, y): 4x^2+y^2 \leqslant 8 \text{ and } y^2 \leqslant 4x\}$ is:

(1) $\pi+4$ (2) $\pi+\frac{2}{3}$ (3) $\frac{\pi}{2}+2$ (4) $\frac{\pi}{2}+\frac{1}{3}$

Q6. JEE Main 2026 (22 January Shift 1)

Let the line $x=-1$ divide the area of the region $\{(x, y): 1+x^2 \leq y \leq 3-x\}$ in the ratio $m: n, \gcd(m, n)=1$. Then $m+n$ is equal to

(1) 27 (2) 25 (3) 28 (4) 26

Q7. JEE Main 2026 (23 January Shift 2)

The area of the region enclosed between the circles $x^2+y^2=4$ and $x^2+(y-2)^2=4$ is:

(1) $\frac{4}{3}(2\pi-3\sqrt{3})$ (2) $\frac{2}{3}(2\pi-3\sqrt{3})$
(3) $\frac{2}{3}(4\pi-3\sqrt{3})$ (4) $\frac{4}{3}(2\pi-\sqrt{3})$

Q8. JEE Main 2026 (21 January Shift 1)

The area of the region, inside the ellipse $x^2+4y^2=4$ and outside the region bounded by the curves $y=|x|-1$ and $y=1-|x|$, is :

(1) $2\pi-1$ (2) $3(\pi-1)$ (3) $2\pi-\frac{1}{2}$ (4) $2(\pi-1)$

Q9. JEE Main 2026 (28 January Shift 1)

The area of the region $\mathrm{R}=\{(x,y): xy\le 8, 1\le y\le x^2, x\ge 0\}$ is

(1) $\frac{2}{3}(24\log_e(2)-7)$ (2) $\frac{1}{3}(40\log_e(2)+27)$

(3) $\frac{2}{3}(20\log_e(2)+9)$ (4) $\frac{1}{3}(49\log_e(2)-15)$

Q10. JEE Main 2026 (24 January Shift 1)

Let A_1 be the bounded area enclosed by the curves $y=x^2+2, x+y=8$ and y-axis that lies in the first quadrant. Let A_2 be the bounded area enclosed by the curves $y=x^2+2, y^2=x, x=2$, and y-axis that lies in the first quadrant. Then $\mathrm{A}_1-\mathrm{A}_2$ is equal to

(1) $\frac{2}{3}(3\sqrt{2}+1)$ (2) $\frac{2}{3}(2\sqrt{2}+1)$

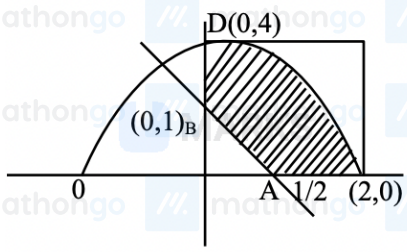
(3) $\frac{2}{3}(\sqrt{2}+1)$ (4) $\frac{2}{3}(4\sqrt{2}+1)$

ANSWERS AND SOLUTIONS

1. (3) **2.** 12 **3.** (1) **4.** (3) **5.** (2) **6.** (1) **7.** (3) **8.** (4)

9. (1) **10.** (2)

1.



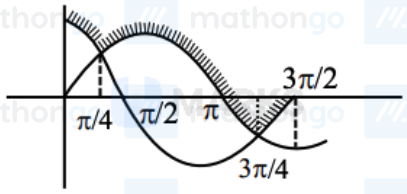
(3)

Required area $= \frac{2}{3} \times 8 - \frac{1}{2} \times \frac{1}{2} \times 1$

$= \frac{16}{3} - \frac{1}{4} = \frac{61}{12} = \frac{\alpha}{\beta}$

$\Rightarrow \alpha + \beta = 73$

2.



(12)

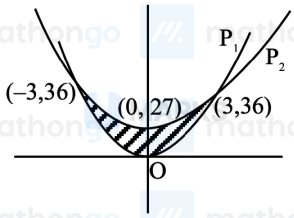
$A = \int_0^{\pi/4} \cos x\, dx + \int_{\pi/4}^{\pi} \sin x\, dx + \int_{\pi}^{5\pi/4} -\sin x\, dx + \int_{5\pi/4}^{3\pi/2} -\cos x\, dx$

$A = (\sin x)_0^{\pi/4} + (-\cos x)_{\pi/4}^{\pi} + (\cos x)_{\pi}^{5\pi/4} + (\sin x)_{5\pi/4}^{3\pi/2}$

$A = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 1 + 1 - \frac{1}{\sqrt{2}} + 1 - \frac{1}{\sqrt{2}} = 3$

$A^2 + A = 12$

3.



(1)

Area bounded between P_1 & P_2 is

$\int_{-3}^{3}((x^2+27)-(4x^2))\, dx$

(P.O.I. of P_1 & P_2 is $x = \pm 3$)

$= 2\int_0^3 (27 - 3x^2)\, dx = 2[27x - x^3]_0^3$

$= 2[81 - 27] = 108$

$\therefore$ Area bounded between P_1 & L is 18 sq. units

(Area between $x^2 = 4ay$ & line $x = my$) is $\frac{8a^2}{3m^3}$

$\therefore$ Area between $x^2 = \dfrac{y}{4}$ & $x = \dfrac{y}{\alpha}$ is

$$\frac{8 \cdot \left(\frac{1}{16}\right)^2}{3 \cdot \left(\frac{1}{\alpha}\right)^3} = 18$$

$$\Rightarrow \frac{\frac{8}{16 \cdot 16}}{\frac{3}{\alpha^3}} = 18 \Rightarrow \alpha^3 = 2^6 \cdot 3^3$$

$$\Rightarrow \alpha = 12$$

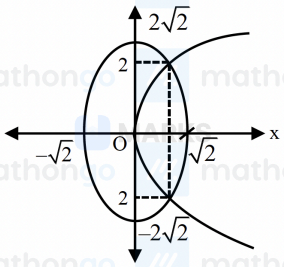
4. **(3)** For $\alpha = 0$: curve is $y = 5 - 1 = 4$ (horizontal line), area $= \int_0^1 (4 - \sqrt{x})dx = 4 - \frac{2}{3} = \frac{10}{3}$

For $\alpha = 1$ and $0 \le x \le 1$: $|x - 5| = 5 - x$, $|1 - x| = 1 - x$, so $y = 4 + x^2$

Area $= \int_0^1 (4 + x^2 - \sqrt{x})dx = 4 + \frac{1}{3} - \frac{2}{3} = \frac{11}{3}$

$f(0) + f(1) = \frac{10}{3} + \frac{11}{3} = 7$

5.



(2)

$$A = \int_0^2 2\sqrt{x}\,dx + 2\int_1^{\sqrt{2}} \sqrt{8 - 4x^2}\,dx$$

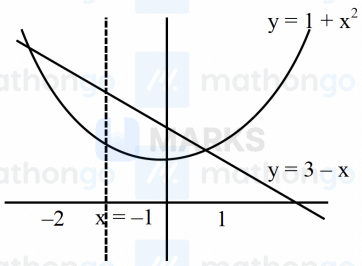
$$= \frac{8}{3}\left(x^{\frac{3}{2}}\right)\Big|_0^1 + 4\int_1^{\sqrt{2}} \sqrt{2 - x^2}\,dx$$

$$= \frac{8}{3} + 4 \times \frac{1}{2}\left[x\sqrt{2 - x^2} + 2\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)\right]\Big|_1^{\sqrt{2}}$$

$$= \frac{8}{3} + 2\left[2 \times \frac{\pi}{2} - 1 - 2 \times \frac{\pi}{4}\right]$$

$$= \frac{8}{3} + 2\pi - 2 - \pi = \pi + \frac{2}{3} \text{ sq. units}$$

6.



(1)

$$\frac{m}{n} = \frac{\int_{-1}^{1}[(3 - x) - (1 + x^2)]\,dx}{\int_{-2}^{1}[(3 - x) - (1 + x^2)]\,dx} = \frac{20}{7}$$

$\therefore m + n = 20 + 7$
$= 27$

7. **(3)** Circle 1: centre $(0, 0)$, $r = 2$. Circle 2: centre $(0, 2)$, $r = 2$. Distance between centres $d = 2$.
Intersection: $y^2 = (y - 2)^2 \Rightarrow y = 1$, giving $x = \pm\sqrt{3}$.
Angle subtended at centre of circle 1 by the chord: $\theta = 2\pi/3$.
Segment area $= \frac{1}{2}(4)\left(\frac{2\pi}{3}\right) - \frac{1}{2}(4)\sin\frac{2\pi}{3} = \frac{4\pi}{3} - \sqrt{3}$.
By symmetry, total intersection area $= 2\left(\frac{4\pi}{3} - \sqrt{3}\right) = \frac{8\pi - 6\sqrt{3}}{3} = \frac{2}{3}(4\pi - 3\sqrt{3})$.

8. **(4)** Ellipse: $\frac{x^2}{4} + y^2 = 1$ with $a = 2$, $b = 1$. Area $= \pi ab = 2\pi$.
Curves $y = |x| - 1$ and $y = 1 - |x|$ intersect at $x = \pm 1$, forming a rhombus with vertices $(0, 1)$, $(1, 0)$, $(0, -1)$, $(-1, 0)$.
Vertices $(0, \pm 1)$ lie on ellipse; $(\pm 1, 0)$ are inside ellipse.
Area of rhombus $= \frac{1}{2} \times 2 \times 2 = 2$.
Required area $= 2\pi - 2 = 2(\pi - 1)$.

9. **(1)** The region R is defined by $xy \leq 8$, $1 \leq y \leq x^2$, and $x \geq 0$.
First, find the intersection points of the curves.
Intersection of $y = 1$ and $y = x^2$: $x^2 = 1 \implies x = 1$ (since $x \geq 0$).
Intersection of $y = x^2$ and $xy = 8$: $x(x^2) = 8 \implies x^3 = 8 \implies x = 2$. At $x = 2$, $y = 4$.
Intersection of $xy = 8$ and $y = 1$: $x(1) = 8 \implies x = 8$.
The region is bounded below by $y = 1$ from $x = 1$ to $x = 8$.
The upper boundary consists of two parts:
From $x = 1$ to $x = 2$, the upper curve is $y = x^2$.
From $x = 2$ to $x = 8$, the upper curve is $y = \frac{8}{x}$.
The area A is given by:
$A = \int_1^2 (x^2 - 1)dx + \int_2^8 (\frac{8}{x} - 1)dx$
$A = [\frac{x^3}{3} - x]_1^2 + [8\log_e x - x]_2^8$
$A = (\frac{8}{3} - 2) - (\frac{1}{3} - 1) + (8\log_e 8 - 8) - (8\log_e 2 - 2)$
$A = (\frac{2}{3}) - (-\frac{2}{3}) + 8(3\log_e 2) - 8 - 8\log_e 2 + 2$
$A = \frac{4}{3} + 24\log_e 2 - 8\log_e 2 - 6$
$A = 16\log_e 2 + \frac{4}{3} - 6$
$A = 16\log_e 2 - \frac{14}{3}$
$A = \frac{2}{3}(24\log_e 2 - 7)$

10. **(2)** For A_1: intersection of $y = x^2 + 2$ and $x + y = 8$ at $x = 2, y = 6$
$A_1 = \int_0^2 (8 - x - (x^2 + 2))dx = \int_0^2 (6 - x - x^2)dx = \left[6x - \frac{x^2}{2} - \frac{x^3}{3}\right]_0^2 = 12 - 2 - \frac{8}{3} = \frac{22}{3}$
For A_2: bounded by $y = x^2 + 2$, $y^2 = x$ (i.e., $x = y^2$), $x = 2$, and y-axis.
In first quadrant, $y^2 = x$ goes from (0,0) to (2,$\sqrt{2}$).

The parabola $y = x^2 + 2$ passes through (0,2) and (2,6).

$A_2 = \int_0^{\sqrt{2}} (2 - y^2) dy = \left[2y - \frac{y^3}{3}\right]_0^{\sqrt{2}} = 2\sqrt{2} - \frac{2\sqrt{2}}{3} = \frac{4\sqrt{2}}{3}$

$A_1 - A_2 = \frac{22}{3} - \frac{4\sqrt{2}}{3} = \frac{2}{3}(2\sqrt{2} + 1)$